ISTA 130
Exercises: Conditional Statements

SOLUTION

1.) Write the result of evaluating each of the following *expressions* in the blank:

a.) **False or True**

```
>>> False or True
True
```

b.) **True and False**

```
>>> True and False
False
```

c.) **not (7 > 3)**

```
>>> not (7 > 3)
False
```

d.) **not (True or False)**

```
>>> not (True or False)
False
```

e.) **(not True) and (not False)**

```
>>> (not True) and (not False)
False
```

2.) Write a Boolean *expression* that evaluates to **True** if the value of the variable **count** is between 1 and 10 inclusive.

```
>>> count = 0
>>> count >= 1 and count <= 10
False
>>> count = 1
>>> count >= 1 and count <= 10
True
>>> count = 2
>>> count >= 1 and count <= 10
True
>>> count = 10
>>> count >= 1 and count <= 10
True
>>> count = 11
>>> count >= 1 and count <= 10
False
```

3.) How does Python know to which **if** an **else** belongs? As for example in this:

```
if name == "Snoopy":
   if color == "white":
      print("Yep, Snoopy")
   else:
      print("Nope, imposter")
```

The indentation.

4.) Write a conditional statement that sets the value of a variable named **answer** to 1 if a variable named **result** is equal to 100. Set **answer** to 2 otherwise.

One way to do this:

```
if result == 100:
   answer = 1
else:
   answer = 2
```

5.) Write a conditional statement that sets the value of a variable **gradepoint** to 4 if a variable named **score** is 90 or above, 3 if **score** is between 80 and 89, 2 if **score** is between 70 and 79, 1 if **score** is between 60 and 69, and 0 otherwise.

One way to do this:

```
if score >= 90:
   gradepoint = 4
elif score >= 80:
   gradepoint = 3
elif score >= 70:
   gradepoint = 2
elif score >= 60:
   gradepoint = 1
else:
   gradepoint = 0
```

6.) A fruit company sells oranges for 32 cents a pound plus $7.50 per order for shipping. If an order is over 100 pounds, shipping cost is reduced by $1.50. Write a function that will take the number of pounds of oranges as a parameter and **return** the cost of the order.

One way to do this:

```
def order_cost(fruit_pounds):
   cost = 7.50 + 0.32 * fruit_pounds
   if fruit_pounds > 100:
      cost -= 1.50
   return cost
```

7.) Write a function that takes three integers as parameters and returns the largest.

One way to do this:

```
def return_largest(a, b, c):
    largest = a
    if b > largest:
        largest = b
    if c > largest:
        largest = c
    return largest
```

8.) Write a function that takes two parameters, a pay rate and the number of hours worked, and returns the pay. Any hours over 40 are paid at time and a half.

One way to do this:

```
def calculate_pay(rate, hours):
    if hours > 40:
        return rate * 40.0 + (hours - 40.0) * (rate * 1.5)
    return rate * 40.0
```

9.) What is the output of the following code?

```
weight_oz = 1.5
length = 28

if weight_oz <= 1.0:
    price = 0.45
elif weight_oz <= 2.0: price
    = 0.65
else:
    price = 1.05

if length > 10:
    price = price + 0.25
if length > 20:
    price = price + 0.50
if length > 30:
    price = price + 1.00
print("Price is: $", str(price))
```

Price is: $ 1.4